## BACHELOR OF SCIENCE IN INFORMATION SYSTEMS PROGRAM

**BS Information Systems -** Capstone Project is required to complete as terminal project requirements that would not only demonstrate a student's comprehensive knowledge of the area of study and research methods used but also allow them to apply the concepts and methods to a specific problem in their area of specification. Students must complete a project such as a business process and the implication of introducing a computing solution to a problem.

## SUGGESTED AREAS FOR THESIS / CAPSTONE PROJECTS

The following is a list of some suggested areas per program. The specific areas identified for each degree program may also be considered for the other computing degree programs, depending on the scope, limitations, and approach and following the principles stated in the preceding sections.

### I. Computer Science

**Current Computer Science Topics**

- Software Development and Theory
- Mobile Computing Systems
- Software Extensions or Plug-ins
- Expert Systems and Decision Support Systems
- Systems Software (software tools/utilities, interpreters, simulators, compilers, security aspects)
- Intelligent Systems
- Game Development
- Computer Vision
- Image / Signal Processing
- Natural Language Processing
- Pattern Recognition and Data Mining
- Bioinformatics
- Graph Applications
- Cloud Computing
- Parallel Computing
- Embedded Systems
- Emerging Technologies

### II. Information Systems

**Current Information Systems Topics**

- Customer Relationship Management (CRM) Application
- E-Learning Platform Development
- Automated Expense Tracking Application
- Health Monitoring Mobile App

- Employee Performance Management System
- Real Estate Property Management Tool
- E-Commerce Website with Analytics
- Event Management System
- Personal Finance Management App
- Social Media Analytics Dashboard
- Online Food Ordering System
- Telemedicine Application for Remote Consultations
- Smart Parking Management System
- Subscription Management Platform
- Collaborative Task Management Tool
- Travel Planning and Booking App
- Virtual Reality Training Application
- Employee Onboarding Portal
- Data Visualization Tool for Business Intelligence
- Fitness Tracker and Wellness App
- Supply Chain Management System
- Digital Marketing Campaign Management Tool
- Customizable Survey and Feedback Application
- Local Community Events Calendar
- Mobile Wallet and Payment Solution
- AI-Powered Chatbot for Customer Support
- Freelancer Marketplace Platform
- Sustainability Tracking Application for Businesses
- Personalized Learning Management System

**Title Defense**

1. Each group will be given a maximum of one hour to present a minimum of three (3) titles for approval. The concept of the study should be discussed to assess the depth and relevance of the proposal. Students should be guided by the “S-M-A-R-T” principles in proposing topics.
2. Two (2) copies of the topic proposal should be submitted to the instructor in charge one (1) week before the scheduled Title Defense.
3. Topic proposal covers the Introduction part of the Thesis / Capstone manuscript
4. The students should be ready to answer questions from the panel members.
5. The panel members will evaluate the presentation using a rubric.
6. Once a title is approved, the group will be assigned an Adviser to guide them through the research process. This approved title will remain until the Thesis Writing 2 / Capstone Project 2 courses are completed.
7. If no titles were approved, the group will prepare three (3) titles again that will be presented after two (2) weeks.

**SAMPLE FLOW OF THE MANUSCRIPT**

**COVER PAGE**
**ABSTRACT**
**INTRODUCTION**
- Background of the Study
- Statement of the Problem
- Objectives of the Study
- Scope of the Study
- Delimitations of the Study
- Conceptual Framework (IPO Diagram)
- Significance of the Study
- Definition of Terms
- Ethical Considerations
- References

**REVIEW OF RELATED LITERATURES AND STUDIES** (5 Years recency and Year today below)
- 5 Foreign Literature
- 5 Local Literature
- 5 Foreign Studies
- 5 Local Studies

**METHODOLOGY**
- Algorithms (BSCS only)
- Research Design (Descriptive & Quantitative Method))
- System Development Methodology
- Participants (Sampling Technique)
- Procedure (such as Interview, Observation, Questionnaire)
- Data Analysis (such as Evaluation Procedures, Computations, and Scales)
- Design and Implementation
- Requirement Specification Analysis (System Modules)
- System Design and Specifications (sample User Interface Output of the system)
- Logical Specifications ( DFD, Use Case, Flowchart..)
- Hardware Specification
- Software Specifications

**RESULTS**
- System / Application Outputs
- Evaluation Results

**DISCUSSION**
- Summary of Findings
- Conclusions
- Recommendations

**REFERENCES**

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**Manuscript's Style and Format**

In writing your scientific work, the following need to be observed:

- Paper: Use high-quality short bond paper (8.5 x 11)
- Font: Times New Roman
- Font Size: 12
- Line & Paragraph Spacing: 2.0
- Text / Paragraph Alignment: Justified
- Margin: Left – 1.5; Right, Top, & Bottom – 1.0
- Page: Top of Page (BSIS – CS Format) and Bottom of Page for IMRAD (Plain Number 3)
- Tables and Figures: (see Appendix A for Table and Figure Formats)
- Drawings / Photos / Diagrams: High-resolution in JPEG (or JPG) or PNG
- Abstract: Unstructured; maximum of 250 words.
- Keywords: 4 to 7 key words
- Citation and Referencing Style – APA 7th Edition
- Use the following website:
    - https://www.citefast.com/?s=APA7
    - https://www.mybib.com/tools/apa-citation-generator
    - https://www.citationmachine.net/apa